clk
rst

exmem_valid_i RTL_MUX

S=1'b0 I0

S=default I1 O S exmem_reg_we0_i RTL_MUX

S=1'b1 I0

S=default I1 O S exmem_valid_reg RTL_REG_SYNC C

CED Q

RST

idex_reg_we_i__0 RTL_MUX

S=1'b0 I0

S=default I1 O S

idex_reg_we_i RTL_MUX

S=1'b0 I0

S=default I1 O S exmem_reg_we_i__0 RTL_MUX

S=1'b0 I0

S=default I1 O S

idex_reg_we_i__1 RTL_MUX

S=1'b1 I0

S=default I1 O S memwb_reg_we0_i RTL_MUX

S=1'b1 I0

S=default I1 O S

idex_rd_i RTL_MUX

S=1'b1 I0

S=default I1 O S

idex_reg_we_reg RTL_REG_SYNC C

CED Q

RST

exmem_reg_we_i__1 RTL_MUX

S=1'b1 I0

S=default I1 O S exmem_reg_we_i RTL_MUX

S=1'b0 I0

S=default I1 O S memwb_reg_we_i__0 RTL_MUX

S=1'b0 I0

S=default I1 O S

idex_rd_i__0 RTL_MUX

S=1'b0 I0

S=default I1 O S

exmem_reg_we_reg RTL_REG_SYNC C

CED Q

RST

idex_rd_reg[2:0] RTL_REG_SYNC C

CED Q

RST

exmem_rd_i RTL_MUX

S=1'b0 I0

S=default I1 O S memwb_reg_we_i__1 RTL_MUX

S=1'b1 I0

S=default I1 O S memwb_reg_we_i RTL_MUX

S=1'b0 I0

S=default I1 O S

exmem_rd_reg[2:0] RTL_REG_SYNC C

CED Q

RSTmemwb_reg_we_reg RTL_REG_SYNC C

CED Q

RST memwb_rd_i RTL_MUX

S=1'b0 I0

S=default I1 O S

id_ra_addr_i RTL_MUX S

S=1'b1 I0[2:0] S=default I1[2:0] O[2:0] exmem_mem_we0_i RTL_MUX

S=1'b1 I0

S=default I1 O S

id_rb_addr_i RTL_MUX S

S=1'b1 I0[2:0] S=default I1[2:0] O[2:0]

memwb_rd_reg[2:0] RTL_REG_SYNC C

CED Q

RST

idex_mem_we_i__0 RTL_MUX

S=1'b0 I0

S=default I1 O S we0_i__0 RTL_AND

I0

I1 O

idex_rb_data_i RTL_MUX

S=1'b1 I0

S=default I1 O S

idex_mem_we_i RTL_MUX

S=1'b0 I0

S=default I1 O S exmem_mem_we_i__0 RTL_MUX

S=1'b0 I0

S=default I1 O S exmem_alu_result_i RTL_MUX

S=1'b0 I0

S=default I1 O S

idex_mem_we_i__1 RTL_MUX

S=1'b1 I0

S=default I1 O Su_regfile

regfile8risc

clk

rst we

ra_addr[2:0] rb_addr[2:0] ra_data[7:0] rb_data[7:0] w_addr[2:0] w_data[7:0]

idex_rb_data_i__0 RTL_MUX

S=1'b0 I0

S=default I1 O S

idex_mem_we_reg RTL_REG_SYNC C

CED Q

RST

exmem_mem_we_i__1 RTL_MUX

S=1'b1 I0

S=default I1 O S exmem_mem_we_i RTL_MUX

S=1'b0 I0

S=default I1 O S

exmem_alu_result_reg[7:0] RTL_REG_SYNC C

CED Q

RST

idex_rb_data_reg[7:0] RTL_REG_SYNC C

CED Q

RST

exmem_mem_to_reg0_i RTL_MUX

S=1'b1 I0

S=default I1 O S

exmem_mem_we_reg RTL_REG_SYNC C

CED Q

RST

exmem_st_data_i RTL_MUX

S=1'b0 I0

S=default I1 O S

exmem_addr_i RTL_MUX

S=1'b0 I0

S=default I1 O S

exmem_mem_to_reg_i__0 RTL_MUX

S=1'b0 I0

S=default I1 O S ex_opb_reg_i RTL_MUX

S=2'b01 I0[7:0] S=2'b10 I1[7:0] S=default I2[7:0] O[7:0] S[1:0]

exmem_addr_reg[7:0] RTL_REG_SYNC C

CED Q

RST

exmem_st_data_reg[7:0] RTL_REG_SYNC C

CED Q

RST

we0_i RTL_AND

I0

I1 O

exmem_mem_to_reg_i__1 RTL_MUX

S=1'b1 I0

S=default I1 O S exmem_mem_to_reg_i RTL_MUX

S=1'b0 I0

S=default I1 O S

u_dmem
dmem8risc

clk

rst we

addr[7:0] rdata[7:0] wdata[7:0]

exmem_mem_to_reg_reg RTL_REG_SYNC C

CED Q

RST

idex_imm8_i RTL_MUX

S=1'b1 I0

S=default I1 O S

idex_ra_data_i RTL_MUX

S=1'b1 I0

S=default I1 O S

idex_alu_src_imm_i RTL_MUX

S=1'b1 I0

S=default I1 O S

mem_result_i RTL_MUX S

S=1'b1 I0[7:0] S=default I1[7:0] O[7:0] memwb_result_i RTL_MUX

S=1'b0 I0

S=default I1 O S

idex_ra_data_i__0 RTL_MUX

S=1'b0 I0

S=default I1 O S

idex_imm8_i__0 RTL_MUX

S=1'b0 I0

S=default I1 O S

idex_alu_src_imm_i__0 RTL_MUX

S=1'b0 I0

S=default I1 O S

idex_alu_op_i RTL_MUX

S=1'b1 I0

S=default I1 O S

memwb_result_reg[7:0] RTL_REG_SYNC C

CED Q

RST

idex_ra_data_reg[7:0] RTL_REG_SYNC C

CED Q

RST

idex_alu_src_imm_reg RTL_REG_SYNC C

CED Q

RST

idex_imm8_reg[7:0] RTL_REG_SYNC C

CED Q

RST

idex_alu_op_i__0 RTL_MUX

S=1'b0 I0

S=default I1 O S

ex_opa_i RTL_MUX
S=2'b01 I0[7:0] S=2'b10 I1[7:0] S=default I2[7:0] O[7:0] S[1:0] ex_opb_i RTL_MUX S

S=1'b1 I0[7:0] S=default I1[7:0] O[7:0]

idex_alu_op_reg[2:0] RTL_REG_SYNC C

CED Q

RST

idex_br_z_i__0 RTL_MUX

S=1'b0 I0

S=default I1 O S

u_alu

alu8risc cz a[7:0] alu_op[2:0] b[7:0] result[7:0]

idex_br_z_i__1 RTL_MUX

S=1'b1 I0

S=default I1 O S

idex_br_z_i RTL_MUX

S=1'b0 I0

S=default I1 O S

idex_br_nz_i__0 RTL_MUX

S=1'b0 I0

S=default I1 O S

idex_jump_i__0 RTL_MUX

S=1'b0 I0

S=default I1 O S

take_branch3_i RTL_MUX

S=1'b1 I0

S=default I1 O S

idex_br_z_reg RTL_REG_SYNC C

CED Q

RST

idex_br_nz_i__1 RTL_MUX

S=1'b1 I0

S=default I1 O S

idex_jump_i__1 RTL_MUX

S=1'b1 I0

S=default I1 O S

idex_br_nz_i RTL_MUX

S=1'b0 I0

S=default I1 O S

idex_jump_i RTL_MUX

S=1'b0 I0

S=default I1 O S

idex_br_nz_reg RTL_REG_SYNC C

CED Q

RST

idex_jump_reg RTL_REG_SYNC C

CED Q

RST

take_branch2_i RTL_AND

I0

I1 O

take_branch2_i__0 RTL_INV

I0 O

halt_in_pipe1_i__0 RTL_AND

I0

I1 O

halt_in_pipe1_i RTL_AND

I0

I1 O

take_branch1_i RTL_OR

I0

I1 O

take_branch1_i__0 RTL_AND

I0

I1 O

halt_in_pipe0_i RTL_OR

I0

I1 O

take_branch0_i RTL_OR

I0

I1 O

halt_in_pipe_i RTL_OR

I0

I1 O

take_branch_i RTL_AND

I0

I1 O

ifid_valid_i RTL_MUX

S=1'b1 I0

S=default I1 O S

ifid_valid_i__0 RTL_MUX

S=1'b0 I0

S=default I1 O S

ifid_valid_i__2 RTL_MUX

S=1'b1 I0

S=default I1 O S

ifid_valid_i__1 RTL_MUX

S=1'b1 I0

S=default I1 O S

ifid_valid_i__3 RTL_MUX

S=1'b0 I0

S=default I1 O S

ifid_valid_reg RTL_REG_SYNC C

CED Q

RST SET

idex_valid2_i RTL_OR

I0

I1 O

idex_valid1_i__0 RTL_INV

I0 O

idex_valid1_i RTL_OR

I0

I1 O

idex_valid0_i RTL_OR

I0

I1 O

idex_ra_addr_i RTL_MUX

S=1'b1 I0

S=default I1 O S

idex_rb_addr_i RTL_MUX

S=1'b1 I0

S=default I1 O S

idex_uses_a_i RTL_MUX

S=1'b1 I0

S=default I1 O S

idex_mem_to_reg_i__0 RTL_MUX

S=1'b0 I0

S=default I1 O S

idex_uses_b_i RTL_MUX

S=1'b1 I0

S=default I1 O S

idex_ra_addr_i__0 RTL_MUX

S=1'b0 I0

S=default I1 O S

idex_mem_to_reg_i RTL_MUX

S=1'b0 I0

S=default I1 O S

idex_rb_addr_i__0 RTL_MUX

S=1'b0 I0

S=default I1 O S

idex_uses_a_i__0 RTL_MUX

S=1'b0 I0

S=default I1 O S

idex_mem_to_reg_i__1 RTL_MUX

S=1'b1 I0

S=default I1 O S

idex_uses_b_i__0 RTL_MUX

S=1'b0 I0

S=default I1 O S

idex_mem_to_reg_reg RTL_REG_SYNC C

CED Q

RST

idex_ra_addr_reg[2:0] RTL_REG_SYNC C

CED Q

RST

idex_rb_addr_reg[2:0] RTL_REG_SYNC C

CED Q

RST

idex_uses_a_reg RTL_REG_SYNC C

CED Q

RST

id_uses_a0_i RTL_AND

I0

I1 O

id_uses_b0_i RTL_AND

I0

I1 O

idex_uses_b_reg RTL_REG_SYNC C

CED Q

RST

u_hazard

hazardu8risc

ex_mem_to_reg
ex_uses_a
ex_uses_b

ex_valid
id_uses_a

id_uses_b mem_mem_to_reg mem_reg_we mem_valid
stall wb_reg_we wb_valid

ex_ra_addr[2:0] ex_rb_addr[2:0] ex_rd[2:0]

fwd_a_sel[1:0]
fwd_b_sel[1:0]

id_ra_addr[2:0]

id_rb_addr[2:0] mem_rd[2:0] wb_rd[2:0]

ifid_instr_i__1 RTL_MUX

S=1'b1 I0

S=default I1 O S

ifid_instr_i__2 RTL_MUX

S=1'b0 I0

S=default I1 O S

ifid_instr_i RTL_MUX

S=1'b1 I0

S=default I1 O S

ifid_instr_i__0 RTL_MUX

S=1'b0 I0

S=default I1 O S

ifid_instr_i__3 RTL_MUX

S=1'b1 I0

S=default I1 O Su_imem
imem8risc

addr[7:0] instr[15:0]

idex_valid_i__0 RTL_MUX

S=1'b0 I0

S=default I1 O S

idex_halt_i__0 RTL_MUX

S=1'b0 I0

S=default I1 O S

idex_flag_we_i__0 RTL_MUX

S=1'b0 I0

S=default I1 O S

ifid_instr_reg[15:0] RTL_REG_SYNC C

CED Q

RST

exmem_halt0_i RTL_MUX

S=1'b1 I0

S=default I1 O S

u_ctrl

ctrlu8risc

alu_src_imm

br_nz br_z
flag_we
halt

jump mem_to_reg mem_we

ra_is_rd

rb_is_rd
reg_we

uses_a

uses_b

alu_op[2:0]
opcode[3:0]

idex_flag_we_i RTL_MUX

S=1'b0 I0

S=default I1 O S

idex_valid_i RTL_MUX

S=1'b0 I0

S=default I1 O S

idex_valid_i__1 RTL_MUX

S=1'b1 I0

S=default I1 O S

idex_valid_i__2 RTL_MUX

S=1'b0 I0

S=default I1 O S

idex_halt_i__1 RTL_MUX

S=1'b1 I0

S=default I1 O S

idex_flag_we_i__1 RTL_MUX

S=1'b1 I0

S=default I1 O S

idex_halt_i RTL_MUX

S=1'b0 I0

S=default I1 O S

exmem_halt_i__0 RTL_MUX

S=1'b0 I0

S=default I1 O S

memwb_halt0_i RTL_MUX

S=1'b1 I0

S=default I1 O S

pc_i__0 RTL_MUX

S=1'b1 I0

S=default I1 O S

idex_flag_we_reg RTL_REG_SYNC C

CED Q

RST

idex_valid_reg RTL_REG_SYNC C

CED Q

RST SET

idex_halt_reg RTL_REG_SYNC C

CED Q

RST

exmem_halt_i__1 RTL_MUX

S=1'b1 I0

S=default I1 O S memwb_halt_i__0 RTL_MUX

S=1'b0 I0

S=default I1 O S exmem_halt_i RTL_MUX

S=1'b0 I0

S=default I1 O S

pc_i__1 RTL_MUX

S=1'b1 I0

S=default I1 O S

take_branch5_i RTL_AND

I0

I1 O

exmem_halt_reg RTL_REG_SYNC C

CED Q

RST memwb_halt_i__1 RTL_MUX

S=1'b1 I0

S=default I1 O S memwb_halt_i RTL_MUX

S=1'b0 I0

S=default I1 O S memwb_valid_i RTL_MUX

S=1'b0 I0

S=default I1 O S+

pc0_i RTL_ADD

I1

I0[7:0] O[7:0] pc_i__2 RTL_MUX

S=1'b1 I0

S=default I1 O S

flag_z_i RTL_MUX

S=1'b0 I0

S=default I1 O S

flag_c_i RTL_MUX

S=1'b0 I0

S=default I1 O S memwb_halt_reg RTL_REG_SYNC C

CED Q

RSTmemwb_valid_reg RTL_REG_SYNC C

CED Q

RST pc_i__3 RTL_MUX

S=1'b0 I0

S=default I1 O Spc_iRTL_MUX S

S=1'b1 I0[7:0] S=default I1[7:0] O[7:0]

flag_z_reg RTL_REG_SYNC C

CED Q

RST

flag_c_reg RTL_REG_SYNC C

CED Q

RST

halted_i RTL_AND

I0

I1 O

pc_reg[7:0] RTL_REG_SYNC C

CED Q

RST

flags_out[1:0]

halted

pc_out[7:0]
instr_done